ATHENS course on Power system flexibility

Optimal Storage Arbitrage under Net Metering using Linear Programming

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Net-metering - a brief overview

What is Net Metering?
... facilitates bi-directional power flow to and from an end-user

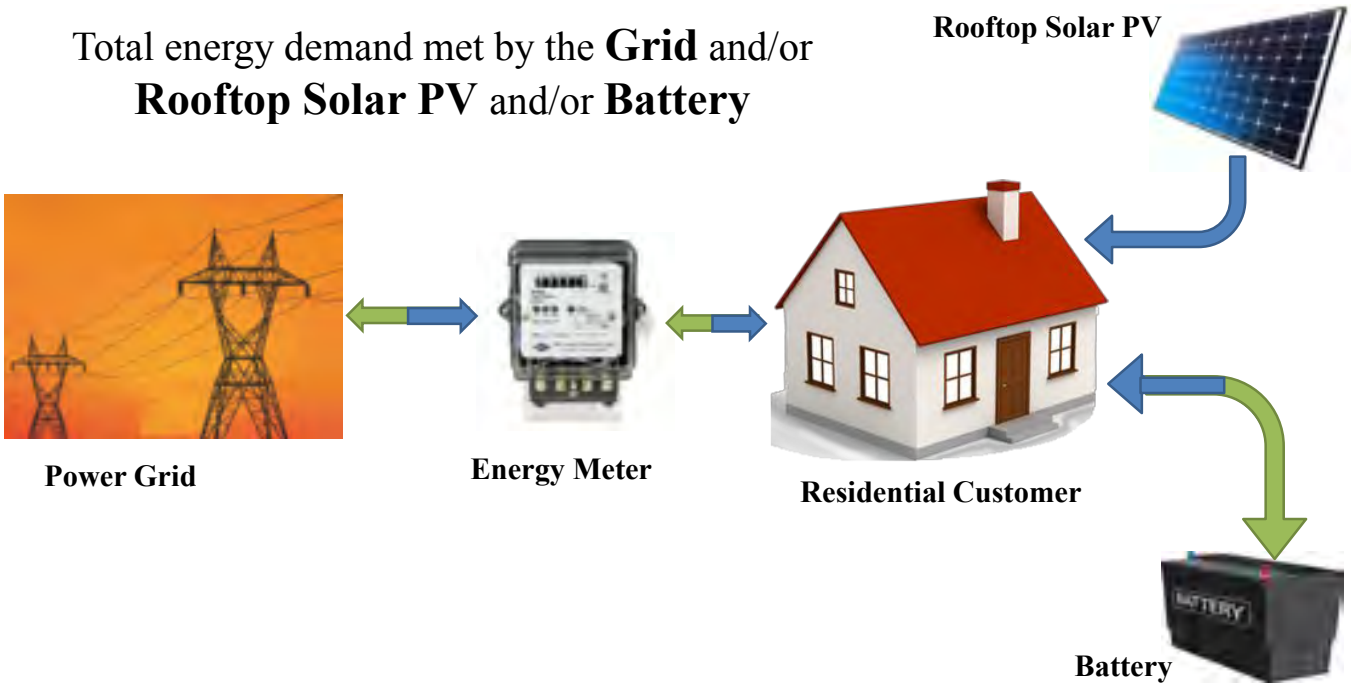
Effect of Net Metering?
... encourage private investment in distributed generation (DG)
... over 200,000 electricity consumers in US generates at least part of their own electricity by 2014

Roadblocks for Net Metering?
... creates disparity for non-DG owners
... could creates localized network congestions cause distributed DG

Compensation mechanisms under Net Metering
... Equal: selling price = retail rate
... Unequal: selling price $<$ retail rate (e.g. avoided cost, generation cost)

System Description: Residential Customer

Total energy demand met by the **Grid** and/or **Rooftop Solar PV** and/or **Battery**



Selection of mode of operation will be based on the price of electricity, solar generation, end user consumption level and battery charge level.

The objective of this work is to **formulate energy arbitrage** under net-metering policies as an LP!!

Reference: Md Umar Hashmi, Arpan Mukhopadhyay, Ana Bušić, and Jocelyne Elias. *Optimal Control of Storage under Time Varying Electricity Prices.* IEEE Smart Grid Communications 2017, Dresden, Germany. Paper Link.

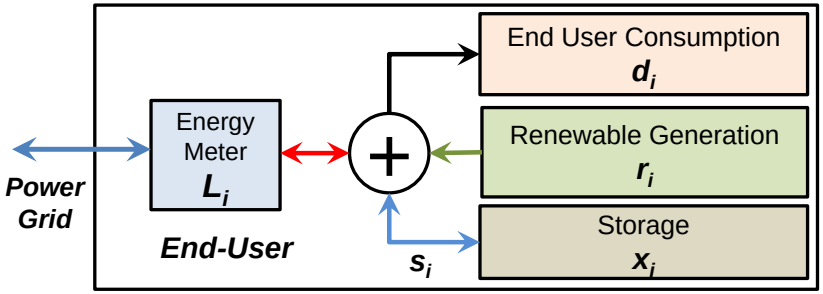
Notation

Electricity price denoted as:

$$p_{\text{elec}}(i) = \begin{cases} p_b(i), & \text{if consumption } \geq 0, \\ p_s(i), & \text{otherwise.} \end{cases} \tag{1}$$

The ratio of selling and buying price of electricity:

$$\kappa_i = \frac{p_s(i)}{p_b(i)} \tag{2}$$



Ramping constraint: $s_i \in [S^i_{\min}, S^i_{\max}]$, where $S^i_{\min} = \eta_{\text{dis}}\delta_{\min}h_i$ and $S^i_{\max} = \frac{\delta_{\max}h_i}{\eta_{\text{ch}}}$.
Capacity constraint: is defined as

$$b_i = b_{i-1} + x_i, \quad b_i \in [b_{\min}, b_{\max}], \forall i. \tag{3}$$

Net-Energy Metering (NEM)

The optimal arbitrage problem is:

$$(P_{\text{NEM}}) \;\; \min \sum_{i=1}^{N} C_{nm}^{i}(x_i),$$

$$\text{subject to, } b_{\min} - b_0 \leq \sum_{j=1}^{i} x_j \leq b_{\max} - b_0, \forall i \in \{1, .., N\},$$

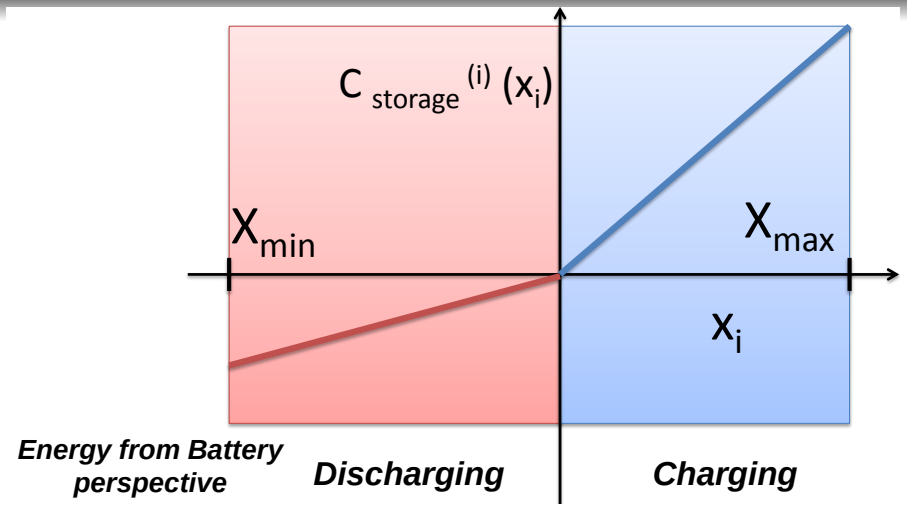
$$x_i \in \left[X_{\min}^{i}, X_{\max}^{i}\right] \forall i \in \{1, .., N\},$$

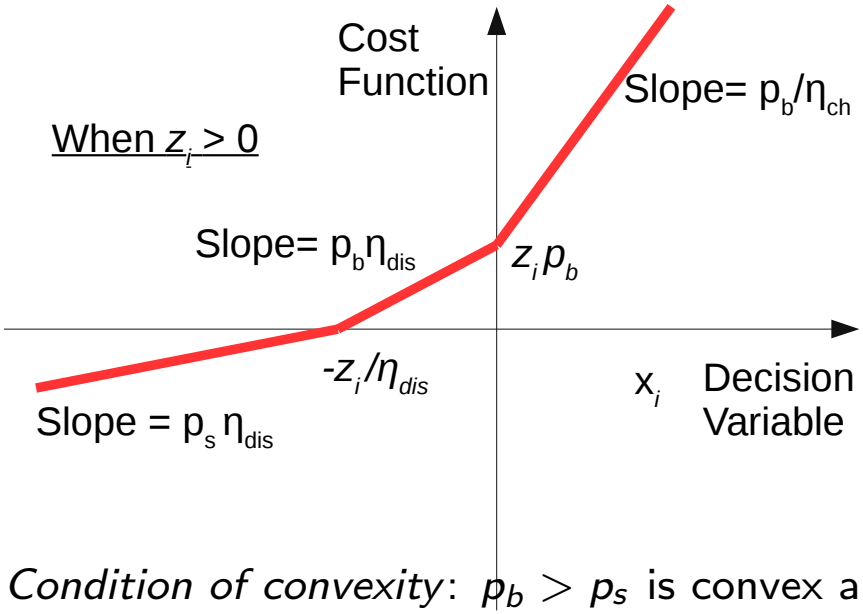
where $C_{\text{nm}}^{i}(x_i)$ denotes the energy consumption cost function at instant i and is given by

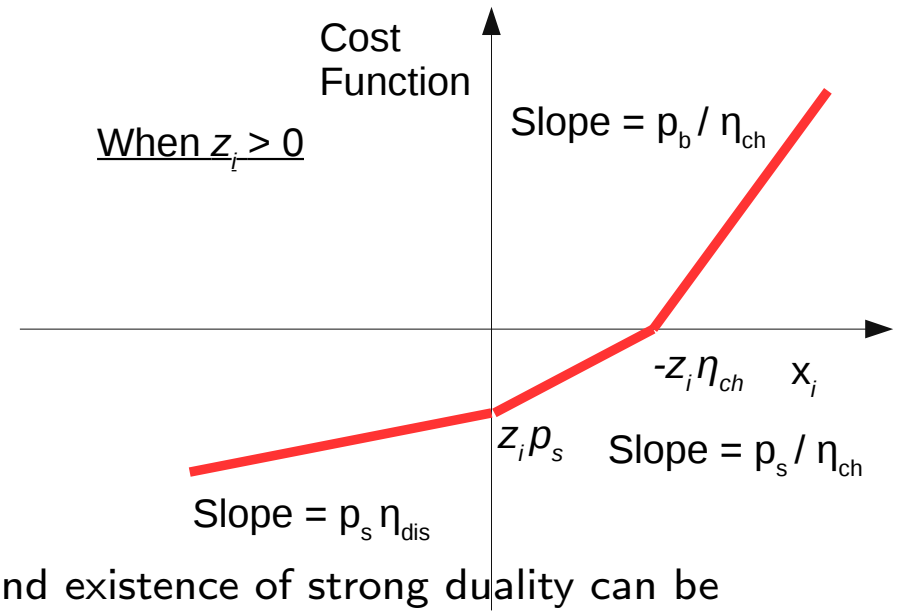
$$C_{\text{nm}}^{i}(x_i) = [z_i + s_i]^{+} p_b(i) - [z_i + s_i]^{-} p_s(i). \tag{4}$$

Reference: • Md Umar Hashmi, Arpan Mukhopadhyay, Ana Bušić, and Jocelyne Elias. *Storage Optimal Control under Net Metering Policies.* To be submitted IEEE Transactions in Smart Grid.
• Md Umar Hashmi, Arpan Mukhopadhyay, Ana Bušić, and Jocelyne Elias. *Storage Optimal Control under Net Metering Policies.* To be submitted IEEE Transactions in Smart Grid. .

Convexity of the cost function







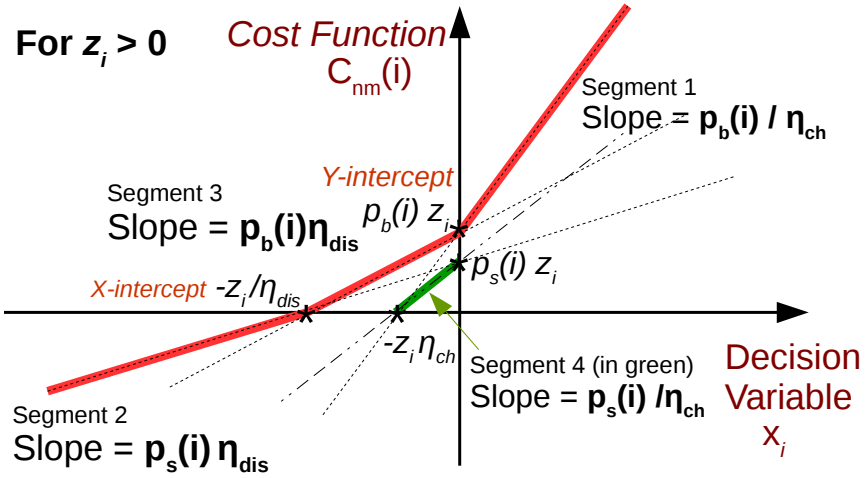
Condition of convexity: $p_b > p_s$ is convex and existence of strong duality can be guaranteed.

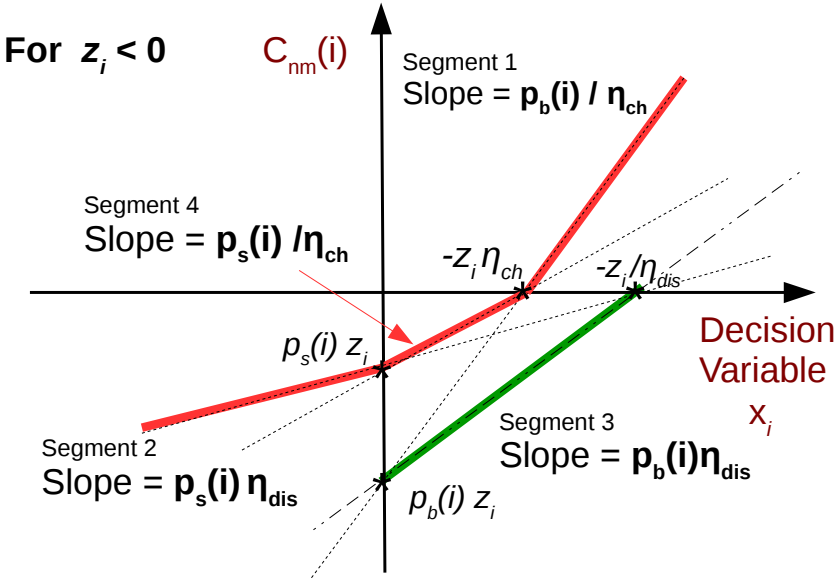
Linear Programming for energy arbitrage

The optimal arbitrage problem, (P_{NEM}), can be solved using linear programming as the cost function is

1. convex
2. piecewise linear
3. associated ramping and capacity constraints are linear

Structure of the cost function





Arbitrage in form of LP based on epigraph

Using the epigraph equivalent formulation for piecewise linear convex cost function we formulate the optimal arbitrage problem using linear programming, denoted as P_{LP}

$$(P_{LP}) \quad \min \quad \{t_1 + t_2 + ... + t_N\},$$

$$\text{subject to,} \quad \text{(a) Segment 1: } \frac{p_b^i}{\eta_{ch}} x_i + z_i p_b^i \leq t_i, \ \forall i$$

$$\text{(b) Segment 2: } p_s^i \eta_{dis} x_i + z_i p_s^i \leq t_i, \ \forall i$$

$$\text{(c) Segment 3: } p_b^i \eta_{dis} x_i + z_i p_b^i \leq t_i, \ \forall i$$

$$\text{(d) Segment 4: } \frac{p_s^i}{\eta_{ch}} x_i + z_i p_s^i \leq t_i, \ \forall i$$

$$\text{(e) Ramp constraint: } x_i \in [X_{\min}^i, X_{\max}^i], \ \forall i$$

$$\text{(f) Capacity constraint: } \sum x_i \in [b_{\min} - b_0, b_{\max} - b_0], \ \forall i.$$

Case 1: Lossless battery without Load

Ojective function: $p_{elec}^T x$, subject to: (a) $A_{c1} x \leq b$, (b) $lb \leq x \leq ub$,

The capacity constraints, $A_{c1} x \leq b$, in matrix is given as

$$\begin{bmatrix} 1 & 0 & 0 & .. & 0 & 0 \\ 1 & 1 & 0 & .. & 0 & 0 \\ : & : & : & .. & : & : \\ 1 & 1 & 1 & .. & 1 & 1 \\ -1 & 0 & 0 & .. & 0 & 0 \\ -1 & -1 & 0 & .. & 0 & 0 \\ : & : & : & .. & : & : \\ -1 & -1 & -1 & .. & -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ : \\ x_N \end{bmatrix} = \begin{bmatrix} b_{\max} - b_0 \\ b_{\max} - b_0 \\ : \\ b_{\max} - b_0 \\ b_0 - b_{\min} \\ b_0 - b_{\min} \\ : \\ b_0 - b_{\min} \end{bmatrix}. \tag{5}$$

The dimension of matrix A_{c1} is 2NxN, and b is of the order 2Nx1.
The lower bound (lb) and upper bound (ub) in constraint (b) is given as

$$lb = \begin{bmatrix} X_{\min} \\ X_{\min} \\ : \\ X_{\min} \end{bmatrix} \leq \begin{bmatrix} x_1 \\ x_2 \\ : \\ x_N \end{bmatrix} \leq ub = \begin{bmatrix} X_{\max} \\ X_{\max} \\ : \\ X_{\max} \end{bmatrix}. \tag{6}$$

Case 2: Lossy battery without Load

$$\min \quad \{t_1 + t_2 + ... + t_N\}$$

$$\text{subject to, } \frac{p_b^i}{\eta_{ch}} x_i - t_i \leq 0, \; p_s^i \eta_{dis} x_i - t_i \leq 0, \forall i$$

$$(i) \; x_i \in [X_{\min}, X_{\max}],$$

$$(ii) \; \sum x_i \in [b_{\min} - b_0, b_{\max} - b_0], \forall i.$$

Matrix format is: minimize $f^T X$, subject to $A_{c2} X \leq b_{c2}$, and $X \in [lb, ub]$.

$$f = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ \vdots \\ 1 \end{bmatrix}, X = \begin{bmatrix} x_1 \\ \vdots \\ x_N \\ t_1 \\ \vdots \\ t_N \end{bmatrix}, lb = \begin{bmatrix} X_{\min} \\ X_{\min} \\ \vdots \\ X_{\min} \\ T_{\min} \\ T_{\min} \\ \vdots \\ T_{\min} \end{bmatrix} \leq \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \\ t_1 \\ t_2 \\ \vdots \\ t_N \end{bmatrix} \leq ub = \begin{bmatrix} X_{\max} \\ X_{\max} \\ \vdots \\ X_{\max} \\ T_{\max} \\ T_{\max} \\ \vdots \\ T_{\max} \end{bmatrix}. \tag{7}$$

where $T_{\min}$ and $T_{\max}$ are bounds on t_i. Since these bounds are not known to us, we choose $T_{\min}$ to be negative with a large magnitude and $T_{\max}$ to be positive with a large magnitude.

Case 2: Lossy battery without Load (Continued)

$$A_{c2} = \begin{bmatrix} \frac{p_b(1)}{\eta_{ch}} & 0 & 0 & \dots & 0 & -1 & 0 & 0 & \dots & 0 \\ 0 & \frac{p_b(2)}{\eta_{ch}} & 0 & \dots & 0 & 0 & -1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \dots & \vdots & \vdots & \vdots & \vdots & \dots & \vdots \\ 0 & 0 & 0 & \dots & \frac{p_b(N)}{\eta_{ch}} & 0 & 0 & 0 & \dots & -1 \\ p_s(1)\eta_{dis} & 0 & 0 & \dots & 0 & -1 & 0 & 0 & \dots & 0 \\ 0 & p_s(2)\eta_{dis} & 0 & \dots & 0 & 0 & -1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \dots & \vdots & \vdots & \vdots & \vdots & \dots & \vdots \\ 0 & 0 & 0 & \dots & p_s(N)\eta_{dis} & 0 & 0 & 0 & \dots & -1 \\ 1 & 0 & 0 & \dots & 0 & 0 & 0 & 0 & \dots & 0 \\ 1 & 1 & 0 & \dots & 0 & 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \dots & \vdots & \vdots & \vdots & \vdots & \dots & \vdots \\ 1 & 1 & 1 & \dots & 1 & 0 & 0 & 0 & \dots & 0 \\ -1 & 0 & 0 & \dots & 0 & 0 & 0 & 0 & \dots & 0 \\ -1 & -1 & 0 & \dots & 0 & 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \dots & \vdots & \vdots & \vdots & \vdots & \dots & \vdots \\ -1 & -1 & -1 & \dots & -1 & 0 & 0 & 0 & \dots & 0 \end{bmatrix}, \quad b_{c2} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 0 \\ 0 \\ \vdots \\ 0 \\ b_{\max} - b_0 \\ b_{\max} - b_0 \\ \vdots \\ b_{\max} - b_0 \\ b_0 - b_{\min} \\ b_0 - b_{\min} \\ \vdots \\ b_0 - b_{\min} \end{bmatrix}, \tag{8}$$

Case 3: Lossy battery with Load and renewable generation

The matrix format for the optimization problem P_{LP} is denoted as minimize $f^T X$, subject to $A_{c3} X \leq b_{c3}$, and $X \in [lb, ub]$.

$$A_{c3} = \begin{bmatrix} \frac{p_b(1)}{\eta_{ch}} & 0 & 0 & \dots & 0 & -1 & 0 & 0 & \dots & 0 \\ 0 & \frac{p_b(2)}{\eta_{ch}} & 0 & \dots & 0 & 0 & -1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \dots & \vdots & \vdots & \vdots & \vdots & \dots & \vdots \\ 0 & 0 & 0 & \dots & \frac{p_b(N)}{\eta_{ch}} & 0 & 0 & 0 & \dots & -1 \\ p_s(1)\eta_{dis} & 0 & 0 & \dots & 0 & -1 & 0 & 0 & \dots & 0 \\ 0 & p_s(2)\eta_{dis} & 0 & \dots & 0 & 0 & -1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \dots & \vdots & \vdots & \vdots & \vdots & \dots & \vdots \\ 0 & 0 & 0 & \dots & p_s(N)\eta_{dis} & 0 & 0 & 0 & \dots & -1 \\ \frac{p_s(1)}{\eta_{ch}} & 0 & 0 & \dots & 0 & -1 & 0 & 0 & \dots & 0 \\ 0 & \frac{p_s(2)}{\eta_{ch}} & 0 & \dots & 0 & 0 & -1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \dots & \vdots & \vdots & \vdots & \vdots & \dots & \vdots \\ 0 & 0 & 0 & \dots & \frac{p_s(N)}{\eta_{ch}} & 0 & 0 & 0 & \dots & -1 \\ p_b(1)\eta_{dis} & 0 & 0 & \dots & 0 & -1 & 0 & 0 & \dots & 0 \\ 0 & p_b(2)\eta_{dis} & 0 & \dots & 0 & 0 & -1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \dots & \vdots & \vdots & \vdots & \vdots & \dots & \vdots \\ 0 & 0 & 0 & \dots & p_b(N)\eta_{dis} & 0 & 0 & 0 & \dots & -1 \\ 1 & 0 & 0 & \dots & 0 & 0 & 0 & 0 & \dots & 0 \\ 1 & 1 & 0 & \dots & 0 & 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \dots & \vdots & \vdots & \vdots & \vdots & \dots & \vdots \\ 1 & 1 & 1 & \dots & 1 & 0 & 0 & 0 & \dots & 0 \\ -1 & 0 & 0 & \dots & 0 & 0 & 0 & 0 & \dots & 0 \\ -1 & -1 & 0 & \dots & 0 & 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \dots & \vdots & \vdots & \vdots & \vdots & \dots & \vdots \\ -1 & -1 & -1 & \dots & -1 & 0 & 0 & 0 & \dots & 0 \end{bmatrix}, \quad b_{c3} = \begin{bmatrix} -z_i p_b(1) \\ -z_i p_b(2) \\ \vdots \\ -z_i p_b(N) \\ -z_i p_s(1) \\ -z_i p_s(2) \\ \vdots \\ -z_i p_s(N) \\ -z_i p_s(1) \\ -z_i p_s(2) \\ \vdots \\ -z_i p_s(N) \\ -z_i p_b(1) \\ -z_i p_b(2) \\ \vdots \\ -z_i p_b(N) \\ b_{\max} - b_0 \\ b_{\max} - b_0 \\ \vdots \\ b_{\max} - b_0 \\ b_0 - b_{\min} \\ b_0 - b_{\min} \\ \vdots \\ b_0 - b_{\min} \end{bmatrix}. \tag{9}$$

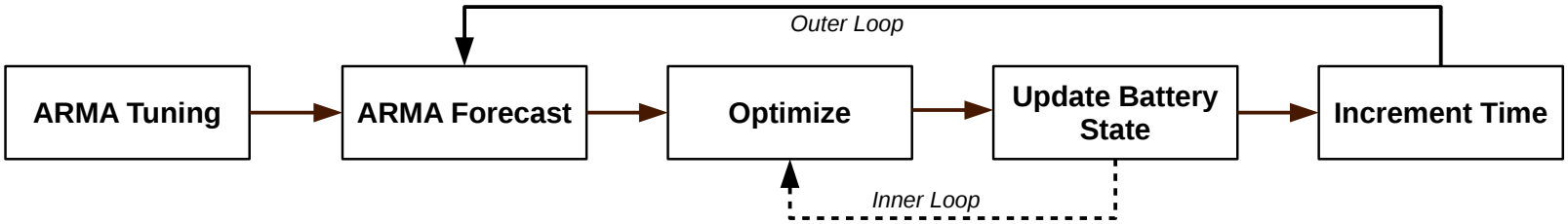
Real time implementation algorithm

`ForecastMPClinearProgram`

Storage Parameters: $\eta_{\text{ch}}, \eta_{\text{dis}}, \delta_{\max}, \delta_{\min}, b_{\max}, b_{\min}, b_0$.

Inputs: $h, N, T, i = 0$, Rolling horizon optimization time period N_{opt}, Historical inelastic load, renewable generation and electricity price data.

Use historical data to tune ARMA and ARIMA models,
while $i < N$ **do**
    Increment $i = i + 1$,
    Real-time electricity price value $p_{\text{elec}}(i)$ and load z_i,
    Forecast $\hat{z}$ from time step $i + 1$ to $i + N_{\text{opt}}$ using ARMA,
    Forecast $\hat{p}_b$ and $\hat{p}_s$ from time $i + 1$ to $i + N_{\text{opt}}$ using ARIMA,
    Calculate $\hat{\kappa}$ as the ratio of $\hat{p}_s$ and $\hat{p}_b$,
    Build LP matrices for time step i to N,
    Solve the Linear Optimization problem for forecast vectors,
    Calculate $b_i{}^* = b_{i-1} + \hat{x}^*(1)$,
    Update $b_0 = b_i{}^*$, the initial capacity of battery is updated.
    Return $b_i{}^*$, $x_i{}^*$.
end while



Storage Degradation Model

Manufacturers define the life of battery based on Calendar and Cycle life. Following are different storage life model:

1. α Cycle degradation[1] $+ \beta$ Ageing degradation[2],
2. f(Cycle degradation, Ageing degradation),
3. max(Cycle degradation, Ageing degradation).

$\downarrow$

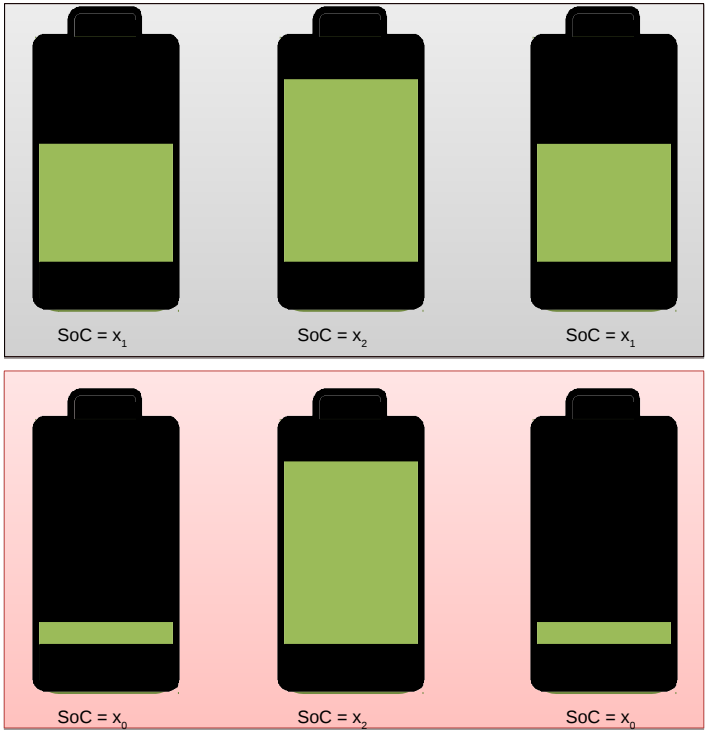
We use Model 3 in this work!!

$\downarrow$

- **Desired condition**: Cycle degradation $\approx$ Ageing degradation, (End-of-life[3] $\approx$ Calendar life)
- **Undesired conditions**:
  - **Over-using**: Cycle degradation $>>$ Ageing degradation, EoL $<<$ Calendar life,
  - **Under-using**: Cycle degradation $<<$ Ageing degradation, EoL $=$ Calendar life.

[1] Cycle degradation refers to battery degradation due to charging and discharging cycles.
[2] Ageing degradation refers to battery degradation due to age of the battery.
[3] End-of-life or EoL is a state where battery capacity reduces to 80% of its initial rated capacity

What is a cycle of operation??



A nonlinear relationship between Cycle Life and Depth-of-Discharge makes it crucial to model degradation of battery.

Dollars per cycle of operation = Total gains / Cycles of operation at x% DoD

Reference: • Md Umar Hashmi, Wael Labidi, Ana Bušić, S.-E. Elayoubi and T. Chahed. *Long-Term Revenue Estimation for Battery Performing Arbitrage and Ancillary Services*. SmartGridComm 2018 at Aalborg, Denmark. Paper Link
• Md Umar Hashmi, Ana Bušić. *Limiting energy storage cycles of operations*. IEEE GreenTech 2018, Austin, USA. Paper Link.

Numerical Experiment Design

Table: Battery Parameters

$b_{\min}, b_{\max}, b_0$	200Wh, 2000 Wh, 1000 Wh
$\eta_{\text{ch}} = \eta_{\text{dis}}$	0.95
$\delta_{\max} = -\delta_{\min}$ (4 battery model)	500 W for 0.25C-0.25C[4], 1000 W for 0.5C-0.5C 2000 W for 1C-1C, 4000 W for 2C-2C

Experiments design:

- Variational analysis of $\kappa_i = p_s^i / p_b^i$.
- Storage parameter variation (Ramp rate)/(Battery Capacity)
- Effect of uncertainty with/without load and renewable generation.

[4] xC-yC implies battery takes 1/x hours to charge and 1/y hours to discharge completely.

Deterministic Simulations for a Day

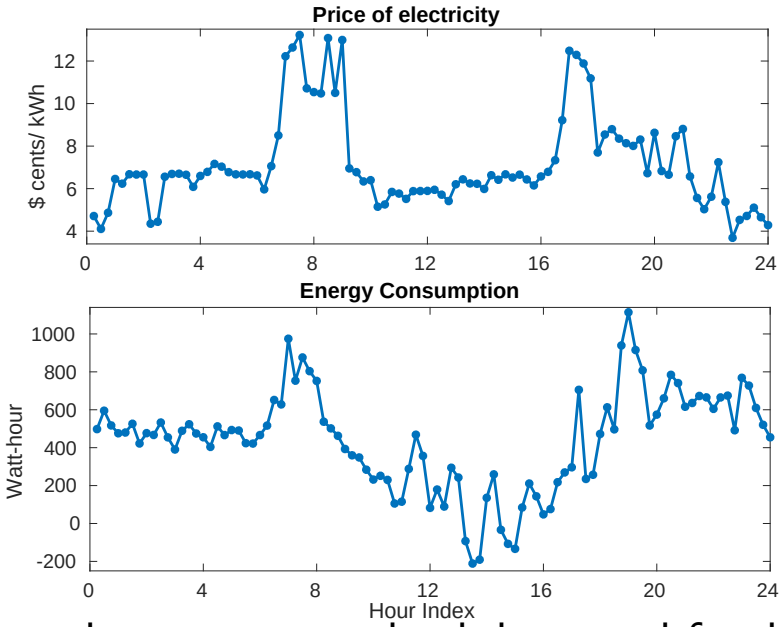


Figure: Electricity price and consumer net load data used for deterministic simulations.

Performance Indices with κ for only storage case

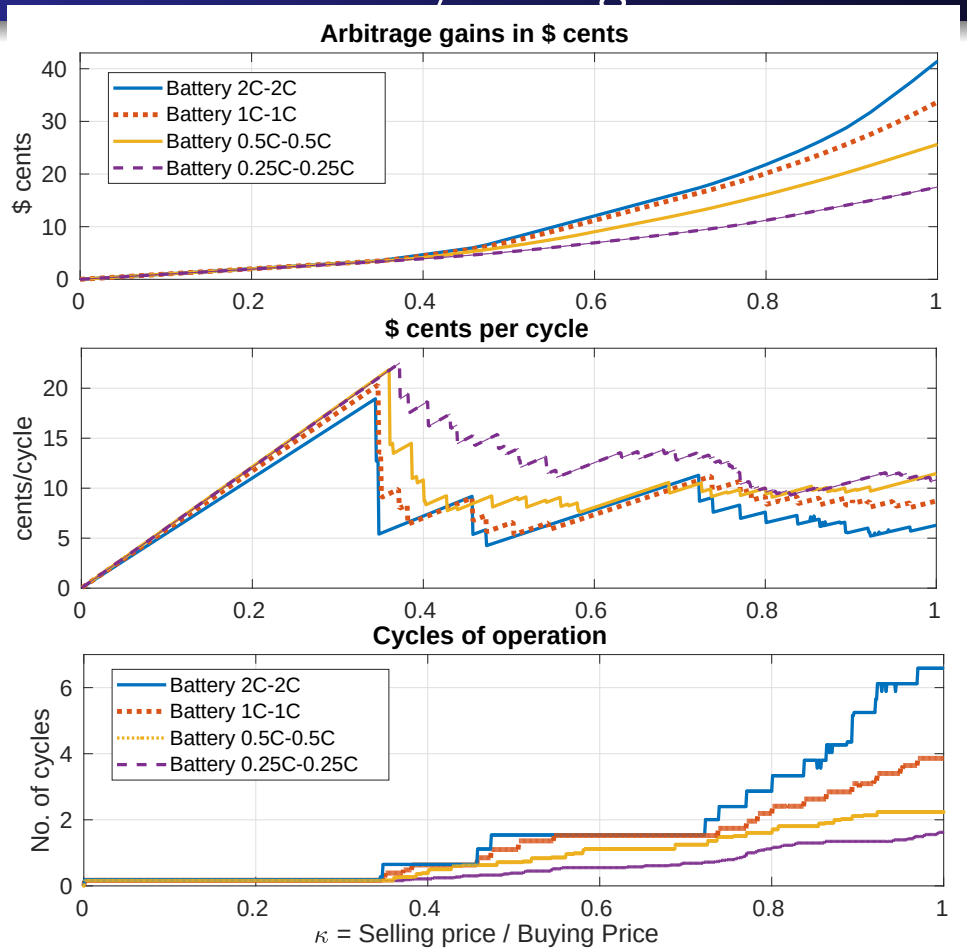


Figure: Performance indices for only storage performing arbitrage with varying κ.

Key observations:

- Storage profitability decreases as selling price decreases

Deterministic results for storage with load and generation

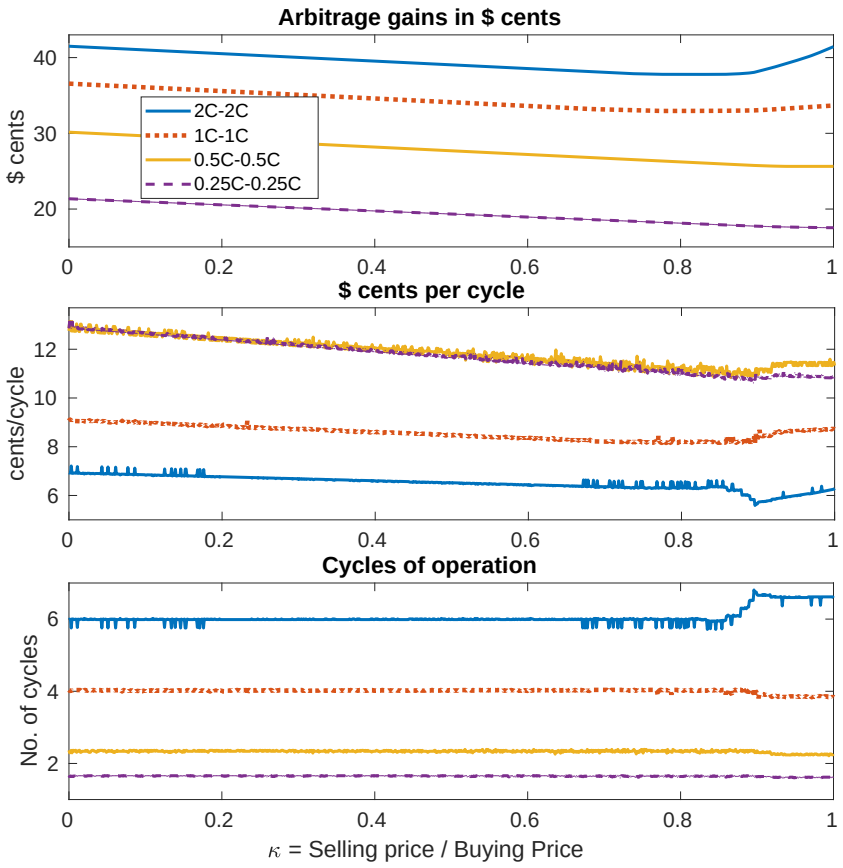


Figure: Storage along with inelastic load and renewable generation with varying κ.

Key observations:

- Consumer load increases storage profitability for lower selling price

Real time implementation of storage with load

Table: Deterministic arbitrage gains for storage with load

κ	2C-2C	1C-1C	0.5C-0.5C	0.25C-0.25C
Arbitrage gains in \$ for 1 week				
1	9.411	7.059	4.784	3.065
0.5	6.641	5.987	4.468	3.019
0	6.313	5.902	4.451	3.019
Cycles of operation for 1 week				
1	58.700	37.294	21.324	12.107
0.5	19.296	16.629	13.007	9.971
0	16.041	15.201	12.484	9.968

Table: Real-time implementation for storage with load

κ	2C-2C	1C-1C	0.5C-0.5C	0.25C-0.25C
Arbitrage gains in \$ for 1 week				
1	6.034	4.684	3.496	3.000
0.5	4.168	3.711	3.292	2.975
0	4.427	3.896	3.396	3.009
Cycles of operation for 1 week				
1	64.322	38.979	22.622	12.850
0.5	34.658	27.627	18.744	11.348
0	32.958	28.255	19.845	11.372

- Faster ramping batteries are more sensitive to parameter uncertainties!!

Conclusion

1. *LP formulation of energy arbitrage* problem with:
   1. consumer load,
   2. renewable generation,
   3. storage capacity and ramping constraint and efficiency losses,
   4. under net-metering with time varying electricity prices.
2. *Real-time* implementation of the LP formulation in context of Model Predictive Control and AutoRegressive Forecasting
3. Numerical experiments indicate:
   1. analyze the sensitivity of energy storage for varying ramp rates and varying ratio of selling and buying price of electricity
   2. the value of storage in presence of load and renewable increases as the ratio of selling and buying price decreases

Open source codes

The LP based optimal arbitrage code described in this paper are publicly available at:

`github.com/umar-hashmi/linearprogrammingarbitrage`

Thank you...

Epigraph based LP formulation

The epigraph problem is denoted as (P_{epi}) $\min t$, subject to, $a_i x + b_i \leq t, \quad i = 1, ..., m,$ where t denotes auxiliary scalar variable. The LP matrix notation for the optimization problem P_{epi} is represented as: minimize $\tilde{f}^T \tilde{x}$, subject to $\tilde{A}\tilde{x} \leq \tilde{b}$; where

$$\tilde{f} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \tilde{x} = \begin{bmatrix} x \\ t \end{bmatrix}, \quad \tilde{A} = \begin{bmatrix} a_1 & -1 \\ \vdots & \vdots \\ a_m & -1 \end{bmatrix}, \quad \tilde{b} = \begin{bmatrix} -b_1 \\ \vdots \\ -b_m \end{bmatrix}.$$

For two time instants the optimization problem is denoted as (P_{epi}) $\min \quad t_1 + t_2$, s.t., (i) $a_{1i}x + b_{1i} \leq t_1$, (ii) $a_{2i}x + b_{2i} \leq t_2$, $i = 1, ..., m$, The equivalent LP matrices are denoted as

$$\tilde{f} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \tilde{x} = \begin{bmatrix} x_1 \\ x_2 \\ t_1 \\ t_2 \end{bmatrix}, \tilde{A} = \begin{bmatrix} a_{11} & 0 & -1 & 0 \\ \vdots & \vdots & \vdots & \vdots \\ a_{1m} & 0 & -1 & 0 \\ 0 & a_{21} & 0 & -1 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & a_{1m} & 0 & -1 \end{bmatrix}, \tilde{b} = \begin{bmatrix} -b_{11} \\ \vdots \\ -b_{1m} \\ -b_{21} \\ \vdots \\ -b_{2m} \end{bmatrix}.$$

A similar LP formulation for N time steps with piecewise linear cost function can be formulated.

Consumer load increases storage profitability for lower selling price

Table: Performance indices for only storage

κ	2C-2C	1C-1C	0.5C-0.5C	0.25C-0.25C
Arbitrage gains in \$ cents for 1 day				
1	44.445	33.760	25.636	17.536
0.5	7.682	7.088	6.253	5.219
0	0	0	0	0
Cycles of operation for 1 day				
1	6.586	3.856	2.237	1.620
0.5	1.539	1.099	0.714	0.386
0	0	0	0	0

Table: Performance indices for storage + load

κ	2C-2C	1C-1C	0.5C-0.5C	0.25C-0.25C
Arbitrage gains in \$ cents for 1 day				
1	44.445	33.760	25.636	17.536
0.5	39.045	34.105	27.696	19.344
0	41.500	36.560	30.150	21.358
Cycles of operation for 1 day				
1	6.586	3.835	2.263	1.620
0.5	5.986	4.033	2.364	1.660
0	5.986	4.033	2.364	1.660

Real time implementation of storage

Table: Deterministic arbitrage gains for only storage

κ	2C-2C	1C-1C	0.5C-0.5C	0.25C-0.25C
Arbitrage gains in \$ for 1 week				
1	9.411	7.059	4.784	3.065
0.75	5.729	4.491	3.168	2.082
0.5	3.166	2.550	1.833	1.217
0.25	1.124	0.941	0.688	0.456
Cycles of operation for 1 week				
1	58.729	37.257	21.324	12.107
0.75	23.462	16.341	10.746	7.519
0.5	12.689	9.770	7.579	6.174
0.25	7.727	6.229	4.558	3.464

Table: Real-time implementation for only storage

κ	2C-2C	1C-1C	0.5C-0.5C	0.25C-0.25C
Arbitrage gains in \$ for 1 week				
1	6.035	4.684	3.469	3.000
0.75	5.024	4.118	3.081	1.904
0.5	3.004	2.367	1.692	1.110
0.25	1.067	0.891	0.618	0.442
Cycles of operation for 1 week				
1	64.323	38.979	22.622	12.850
0.75	24.870	16.169	10.570	7.733
0.5	11.393	8.891	7.013	6.099
0.25	6.429	5.557	4.359	3.395